



PRACTICAL CENTRE

CLASS-XII

BIOLOGY

CHAPTER # 19

ANIMAL BEHAVIOUR

2023-24



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ANIMAL BEHAVIOUR

19

KEY UNITS

THE NATURE OF BEHAVIOUR

INNATE BEHAVIOUR

LEARNING BEHAVIOUR

SOCIAL BEHAVIOUR

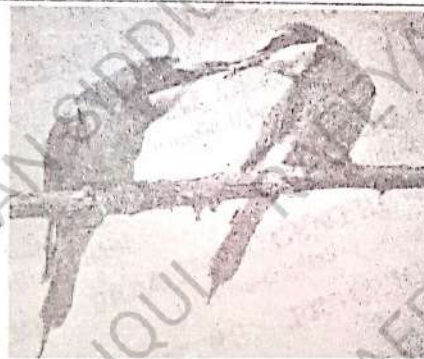


Fig. 19.1

INTRODUCTION:

The Living organisms have the capacity to response against any change in the environment. The response must be adoptive to survive and reproduce. It is the behaviour of an organism which changes in response to any stimulus. Stimulus is anything which brings about change. The origin of stimulus may be outside or inside an organism. The behaviour of an organism is inherited but it can be modified with the passage of time and experience.

Definition:

Behaviour is defined as the aggregates of responses for some stimuli for a given situation over a period of time.

THE NATURE OF BEHAVIOUR:

Behaviour of Organism in response to stimuli.

What an organism does, and how it acts, is collectively known as behaviour. The response of an organism against any stimulus varies from very simple to more complex one. Like responding the aroma of food is a simple response whereas fixed action pattern is a complex behaviour.

The study of behaviour in animals is called *Ethology*. (Gr. ethiologica meaning depicting)



Fig. 19.2

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Relation between Stimulus and Behaviour:

The stimulus brings any detectable change which may be physical or chemical. It must have occurred within environment and enables an organism to response. The body is covered with numerous receptors which detect any change in the surrounding and change it into nerve impulse which will analyze by integrator or sensory organs and then the decision is made whether to respond the signals or it may be ignored.

Interpretations and Response to Stimuli:

The receptors detect any change inside or outside the body. They send message to the CNS through sensory nerve. CNS also known as integrator can receive, gather and analyze the messages. CNS give orders to the effectors through motor nerve.

For example, Reflex action.

Different Ways to Response Stimuli:

The living organism may response to any stimulus in three ways. It can be positive, negative or ignored. In a *positive* response your pet become happy to see the meal. In a *negative* response pigeons close their eyes to see a threat. A child *ignores* you if you show anger on his/her any action.



Fig. 19.3

Relation between Heredity and Behaviour:

The nervous and chemical response can influence the behaviour of an organism. Actually, genes are responsible for any behavioral response because genes develop nervous system. Behaviour is also depended upon secretions of various hormones which are made up of proteins. Proteins are made up of amino acid which is coded by genes.

The animals can exhibit various types of behaviour due to variations in their genes. Artificial selection brings various genes due to selective interbreeding which have different response to a same stimulus. e.g. Dogs, Pigeons, Parrots etc.



Fig. 19.4

BIOLOGICAL RYTHMS:

Bio-rhythm are the physiological frequent pattern of aggregate activities, in an organism against the episodic changes which occur in the environment.

In a living system, the mechanism, which controls the physiological activities internally, without the influence of external signal, is called *Biological Clock*.

If a bio-rhythm appears repeatedly for a specific period of day or 24 hours called *Circadian Rhythms*. Some animals have annual rhythms or *Circannual Rhythms* like migration of bird, hibernation and matting.

- ❖ Either their timing is controlled externally (Exogenous rhythms) or there is internal timing (Endogenous rhythms). Exogenous rhythms are controlled by external changes, such as 24 hours cycle of day and night.
- ❖ Endogenous rhythms are controlled by biochemical and physiological changes, within the organisms. In a number of organisms, it seems a blend of both exogenous and endogenous rhythms. Some examples of rhythms are given below.

Those living organisms which are active during daylight are classified as *Diurnal* animals. Animals hunt, graze or feed during daylight and fall asleep at night. Their eyesights are better in daylight. Sunlight plays a vital role to drive these animals about their routine. *For example:* Butterflies, bird and mammals, including man.

Those living organisms which are active at night are called *Nocturnal* animals. They search for their food in the dark. Their eyes are usually large and unable to watch clearly in bright light. They are not dependent upon sunlight for their routine. *For example:* Cockroach, Owls, Bats etc.

Those living organisms, which are active throughout the day, are grouped as *Crepuscular* animals. They are active in early morning (Dawn) or in early evening (Dusk). They can utilize both types of light i.e. bright and dull more efficiently. *For example:* Mosquito.



Fig. 19.5

TYPES OF BEHAVIOUR:

Nearly all the behaviours, we observe in animals, are adaptive. Animal behaviour can be explained in two ways:

- I) INNATE (INSTINCT) BEHAVIOUR
- II) LEARNING BEHAVIOUR
- III) SOCIAL BEHAVIOUR

(I) INNATE BEHAVIOUR:

Innate behaviour involves automatic pre-programmed, genetically determined, stereotype activities which do not involve any learning. A fish has inherited power of

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swimming by birth. A bird has inherited a power of flying, these are examples of innate behaviour.



Fig. 19.6

There are two types of innate behaviour:

- A) ORIENTATION BEHAVIOUR
- B) NON-ORIENTATION BEHAVIOUR

A) ORIENTATION BEHAVIOUR:

In this kind of behaviour, an organism shows movement and changes its direction in response to a source of stimuli.

It has following two types:

- a. KINESIS b. TAXIS

a. Kinesis:

This behaviour is related to the direction of stimulus. It may be towards or away from the stimulus. The movement towards the stimulus is called positive taxis while the one away from the stimulus is called negative taxis. A moth moves towards light. This is positive phototaxis. It may be phototaxis in response of light, chemotaxis in response to chemical, hydro taxis due to water or thigmotaxis due to touch.



Fig. 19.6a

b. Taxis:

In this behaviour, an organism changes the speed of random movements which help them to survive in the environment. It is a type of orientation movement. The rate of movement is dependent upon the intensity of movement. e.g. wood-lice move about quickly in dry conditions but slow down and stop in humid areas.

B) NON-ORIENTATION BEHAVIOUR

In this type of behaviour, as a result of any stimulus, animals do not show movement in a particular direction. On the basis of complexity, these behaviours are more profound than orientation behaviour. The common types are reflexes, instincts and motivations etc.

a) REFLEXES:

It is the simplest type of behaviour. It is the movement of body parts, such as knee-jerks, blinking of eyes, and withdrawal of hand from a hot object. These are stereotyped, short-lived, rapid responses, mediated by nervous system. During knee jerk, the spinal cord stimulates and through motor nerve, contracts extensor muscles which result in kicking action. Sometimes, the whole body of an animal is involved in reflex action. This type of reflex action is exhibited by invertebrates (worms, snails etc.). This will happen when a nerve fibre carries a very fast message to all muscles, as a result, they contract rapidly and instantaneously.

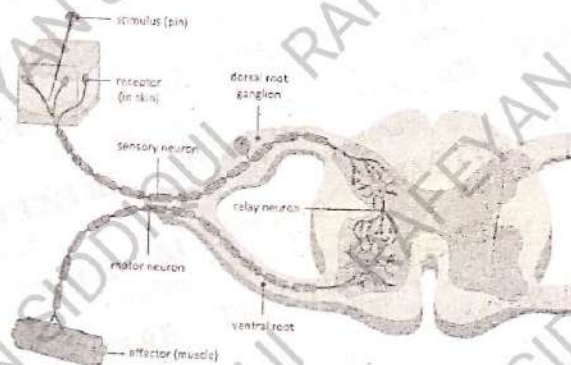


Fig. 19.7

b) INSTINCTIVE BEHAVIOUR:

It is a kind of innate behaviour which belongs to non-orientation behaviour. It is a built-in and stereotype-inherited behaviour pattern, which is not modified by learning or practice. In this behaviour, the whole body of the animal is involved and shows its own characters.

It is stereotype, genetically programmed complex behaviour which evolves gradually and develops side by side with the development of nervous system. It is also called fixed action pattern.

Following are the types of instinctive behaviour:

i. Migration of Salmon fish:

Migration of salmon is tremendous. Salmon are born in fresh water. Female deposits her eggs in fresh water and male sheds his sperms on her eggs which is called *Spawning*. When eggs are hatched, the young spend their early life in fresh water then they move to deep oceans, thousands of miles away. Here, they feed and grow for six months to seven years. After two or three years, salmon transforms into *Smolt* (sexually mature).

During breeding season, these smolts stop feeding and move towards the fresh water spring where they were born. The journey starts from oceanic feeding ground to hundreds of miles away from their spawning grounds where they hatched.

This will risk their life with exhaustion or by predation, only a small group can reach there. This migration is a good example of instinct behaviour.

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ii. Dances of Bee:

Honeybees show very interesting behaviour of communication system to convey signals to other bees to indicate distance and direction of food. Bees (scout) give information to other bees from the hive about the nectar.

The bees perform two types of dance:

- Round Dance
- Waggle Dance.

If the source of food is within 90 metres from the hive, the bees perform round dance. If a source is beyond 90 m, the bees perform waggle dance.

This will indicate the proper direction and distance. The dance may be repeated several times and the sun serves as compass.

This type of communication is the best example of innate behaviour.

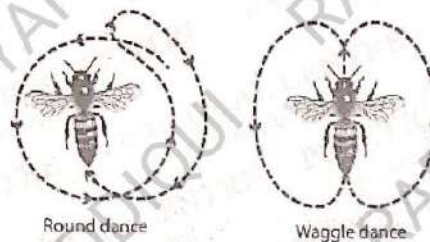


Fig. 19.8

iii. Nest Building:

This behaviour is also categorized in Innate behaviour. Different birds have different strategies to build their nest. For instance a long tailed tailor-bird builds a nest for its mate to lay eggs. Its nest is between the leaves and twigs of the plant and hangs on the soft branches. The tailor bird does not learn nesting from older birds and has no prior experience. It builds nest by its innate behaviour.

iv. Building of Spider's orb web:

The circular or orb web by common spider is a very common web. It is made up of silk thread which is secreted by special silk glands which are present in the abdomen of the spider. The silk is made up of scleroprotein which becomes hard on exposure to air.

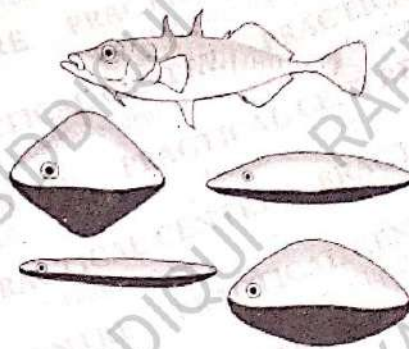
Spider follows a main rule and then the sub rules. By following the program of primary rule, the spider can build a web without having a plan for it. It is an example of highly complex instinct behaviour.

v. Courtship behaviour of Stickleback fish:

This behaviour is triggered or released by an external sensory stimulus known as sign stimulus or releaser. The example is found in male three spined stickleback fish which develop on sexual maturity-bright throat and red belly called nuptial coloration. When the breeding season starts the territorial male builds a nest.

Male becomes aggressive and dominant to protect its nest from other males and attacks other males that enter his territory. But when a female with silver belly enters, the male welcomes it and stimulates spawning by nipping the base of her tail. After spawning, the

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(II) LEARNING BEHAVIOUR:

Fig. 19.10

- i. Habituation
- ii. Imprinting
- iii. Conditioning (Classic conditioning)
- iv. Trial And Error (Operant conditioning)
- v. Latent Learning
- vi. Insight Learning

(i) Habituation:

This habituation is significant and saves energy and time.

(ii) Imprinting:

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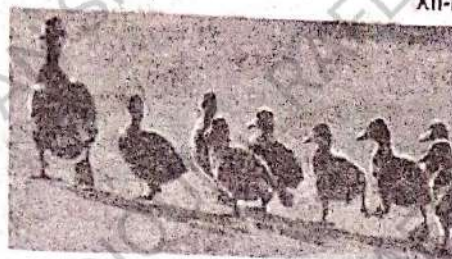


Fig. 19.11

(iii) Classic Condition:

This type of behaviour is associated with some reward or punishment. Ivan Pavlov, in 1902, performed an experiment, using a dog. He showed food to a hungry dog at a particular time with a bell ring. The mouth of the dog started salivation. After some period, he stopped the food but on ringing a bell alone, the dog's mouth started salivation. The response of salivation was first associated with food but now it was associated with bell. A new stimulus which had a meaning for the dog.



Fig. 19.12

(iv) Operant Condition (Trial and error):

In this kind of behaviour, the animal learns something by an award or punishment. An American psychologist, B.F. Skinner, developed a Skinner box for his experiments with hungry pigeons or mice on conditioning. Inside the box, there was a lever with operated food supply. When the lever was pressed by the animal, a sample of food was delivered. After several accidents of pressing the lever, the hungry animals learned to press the lever at the time of their hunger.



Fig. 19.13

(v) Latent Learning:

This type of learning is not associated with some reward or punishment. The ability of rats to find their way in underground tunnels is very remarkable. If a rat is placed in a maze, it runs and keeps on wandering to find the end without being awarded. This type of learning is useful when a predator pursues them. Only then, the rats make use of their latent learning.

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III. SOCIAL LEARNING:

Aggregation:

Animal Society:

A large group of seals, likely fur seals, are resting on a sandy beach. They are mostly light-colored with darker faces and are scattered across the frame, some sitting up and others lying down. The background is a vast, open beach with a line of trees in the distance.

Fig. 19.16

Hostile and Helpful Interspecific Interaction:

In many social insects, hostile and helpful interactions are observed side by side. The best example is seen in honey bees. In their hive, there are three castes of honeybees i.e. workers, drones and queen. It looks hostile when a worker is getting old, her coworker killed her. But it is helpful that every caste works together, for instance, workers collect the nectar and made honey, drones protect the hive and fertilize the eggs and queen lays the eggs.

Agonistic Behaviour:

In the animal world, due to limited resources, animals are usually involved in conflicts due to limited resources of food, mates or territory. These conflicts are solved by the *Agonistic behaviour*. In this behaviour, one animal attacks other animal or becomes aggressive including threats and sometimes combat to determine the right of mating or access to resources. Sometimes, male indicates signals to warn other competitors. Agonistic behaviour decides the territory and dominant hierarchy.

Territory:

A territory is defined as a confined area which an animal reserves for feeding, mating and raising young ones. This area is established by agonistic behaviour by a male to avoid interference or competition from other males.

Once the territory marked by various methods like pug marks, scratching, pee or dung etc. other animals will not enter into it. If they enter, it will be taken as a declared war against the territory male.

Types of Territorial Behaviour:

Animals from insects to worm or fish or birds to mammals show very aggressive behaviour regarding their territory.

Following are some examples of territorial behaviour in various animals:

a. Territorial Behaviour in Gorillas:

Gorilla is non territorial social mammal lives in a group commonly called as troop. A troop may have 1 to 4 old and strong male called silverback. Some juvenile male, are also present called black back. And several adult and immature females.

The silverback is a dominant alpha male of a troop who makes all troops decisions and get larger. He has the right for mate. Silverback is more aggressive than other troops members because the safety of troop is his responsibility. He exhibits aggressive territorial behaviour by standing on his two legs and beat profoundly his chest address the threat.

The adolescent female transfer from other troop into new troop before reproducing age which is 8 years in gorillas can get her rank on the basis of her recruitment order in her new troop. Late arrivals are not supposed to get the benefits of high-ranking females.

On the other hand, adolescent male departs from their families remain alone until they form their own family.



Fig. 19.17

b. Territorial Behaviour in Baboons:

Baboons are dimorphic i.e. their males are larger than their females. Males are territorial and aggressive. They frighten other by showing their large sharp canines. Baboons travel throughout the day for food and rest on trees. The alpha male guides the troop to the feeding ground, water holes and safer trees for rest.

In baboon family, to be a dominant, is not an easy task. It is the responsibility of dominant male to fight predators and invaders which may cause severe injury or death. In baboon, vocalization and displays is enough to defend a territory instead of fighting.

Dominance Hierarchy:

It is defined as the situation in which the animal determines a rank which decides the access to resources like food, mating and grooming services for other members of a social family. Individuals on top of hierarchy have the right for mating, excess food and safe territories. This behaviour takes place due to the exhibition of aggressive behaviour and due to the response to aggression.

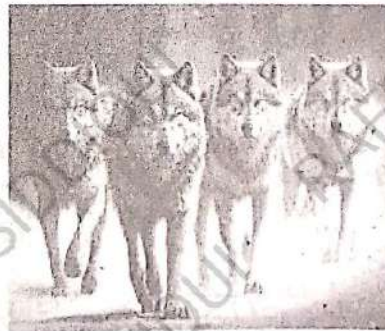


Fig. 19.18

i. Pecking Orders of Chicken:

Dominance hierarchy can be observed in chickens as a good example. When several unfamiliar hens are placed together, they response by chasing, fighting and pecking among themselves until there is established a prominent pecking order. The alpha hen has the right to access resources first like food, water and roosting sites. The alpha hen usually drives off all the other hens by threats instead of actual pecking. The beta or second rank hen reduces all other hens except alpha hen and gradually lower down the rank to the omega. Once the hierarchy is set, a peaceful co-existence is possible.



Fig. 19.19

Altruism:

In animal world, many social behaviours are selfish in which they behave for their own benefit. This kind of behaviour increases the rate of survival and success of reproductive outcome at the expense of others. On the other hand, this type of behaviour is performed by animals without showing self-interest. Animals withdraw their own reproductive potential to rewards other animals. In social animals like insects, the reproduction is confined to only one pair i.e. the queen and her mate.

Altruism in the Organization of Honey Bee:

Honeybees are social insect which live together in a colony or group in an organized manner. The colony has three types of bees castes i.e. the queen, the workers and the drones.

The queen lays fertilized and unfertilized eggs which develop into other bees. New queen develops from fertilized egg. The larva of the queen feed upon special food called Royal jelly.

Both old and new queen simultaneously develop in the same hive for a short time.

Sometimes, the emerging queen may be killed by the other members of hive and the business will run as usual. Or the emerging queen may create a group called swarm and leave the hive to establish a new hive.

On the other hand, sometimes the old queen moves from her hive with her swarm and develops a new hive or may killed, lost or removed and the new queen becomes the owner of hive.

The drones are male bees which develop from unfertilized egg and their function is to fertilize the unfertilized eggs of queen. The mating take place during flight. Drones die just after mating. The workers are sterile females developed from fertilizes eggs. Workers are in abundance but smaller in size. The workers can do all the duties like cleaning hive, formation of beeswax, forage for hive, care and protection for the queen and hive. Although the workers spend their lives in serving and feeding other members of hive, they are sterile.

Honey Bee Caste



Fig. 19.20